

GPS SPEEDOMETER For Cars And Bicycles

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In today's fast-paced world, accurate speed measurement is crucial across various domains, from automotive vehicles to drones and athletic training. To meet this need, the following is a ground-breaking development: a GPS-based speedometer capable of measuring speeds up to 1854 kilometres per hour. What sets this speedometer apart is its compact size, utilising the Quectel L86 GPS + GPS module and the IndusBoard with the 3D accelerometer and compass sensor, all packaged in a tiny 3cm IndusBoard coin form factor.

This makes it the smallest speedometer device, suitable for any automobile, bicycle, runner, athlete, speed bot, motor bot, drone, or robot, enabling real-time speed measurement. In addition, the device also provides IoT capability and connectivity to almost any Wi-Fi enabled device like a laptop, phone, or tablet to keep real-time track of speed and total distance covered.

The applications of this GPS-based speedometer are diverse and impactful. It can be used in automobiles to provide real-time speed feedback for drivers, enhanc-

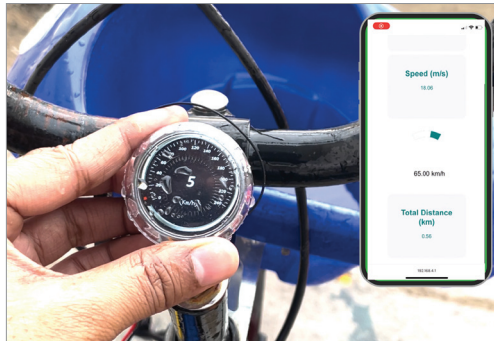


Fig. 1: Author's GPS speedometer for bicycle

ing safety and adherence to speed limits. It can be used in bicycles to help cyclists monitor their speed during training or recreational purposes. It is also beneficial for runners and athletes to provide precise speed measurements for performance tracking and training optimisation. Moreover, it facilitates speed control and navigation accuracy in

```
Speedometeriot.ino  NotoSansBold36.h  dial.h
1
2 #define NEEDLE_LENGTH 8 // Visible length
3 #define NEEDLE_WIDTH 8 // Width of needle - make it an odd number
4 #define NEEDLE_RADIUS 118 // Radius at tip
5 #define NEEDLE_COLOR1 TFT_MAROON // Needle periphery colour
6 #define NEEDLE_COLOR2 TFT_RED // Needle centre colour
7 #define DIAL_CENTRE_X 120
8 #define DIAL_CENTRE_Y 120
9 #include <TinyGPS++.h>
10 #include <HardwareSerial.h>
11 #include <WiFi.h>
12 #include <WebServer.h>
13 #include <Arduino_JSON.h>
14 #include <math.h>
15
16 const char* ssid = "ESP32_S2_AP"; // SSID for the Access Point
17 const char* password = "password"; // Password for the Access Point
18 const int serverPort = 80;
19 float totalDistance = 0.0;
20 WebServer server(serverPort);
21
22 // The TinyGPS++ object
23 TinyGPSPlus gps;
24
25 // Use HardwareSerial for GPS
26 // On ESP32, Serial1 can be repurposed to use custom pins.
27 // RX (GPS TX) is 43, TX (GPS RX) is 44
28 HardwareSerial GPSSerial(1);
29
30 // Font attached to this sketch
31 #include "NotoSansBold36.h"
32 #define AA_FONT_LARGE NotoSansBold36
33
34 #include <TFT_eSPI.h>
```

Fig. 2: Snippet of the source code

BILL OF MATERIALS

Components	Description
IndusBoard	3cm sized development board
L86 GPS module	GPS module
GC9A01A round display	LCD display
CST816	Capacitance touchchip
Casing	Outer casing of device